

scenario, DO NOT uninstall, but follow the procedure below which will help you to build Glassfish as a cluster over the standalone build.

Consider this to be the First Node:
GF-Node01 192.168.10.10

The added directories under `/appinstall/glassfish`, after building a standalone application server are:

- bin
- config
- domains

Delete the above mentioned directories. The reference command is `# rm -rf bin/ config/ domains/`

Set Executable Permission to 'lib/ant/bin' modules under the 'glassfish' directory.

The reference command is `# chmod -R +x lib/ant/bin/`. Ideally, it will be set since the server was previously built as a standalone application server.

Using the 'ant' executable (located under `/appinstall/glassfish/lib/ant/bin`), run `setup-cluster.xml` (located under `/appinstall/glassfish`) to build a Glassfish cluster.

The reference command is `# lib/ant/bin/ant -f setup-cluster.xml`

As mentioned in my last article, under the Glassfish directory, there are two Glassfish `setup.xml` files:

- `setup.xml`: For building a standalone Glassfish environment.
- `setup-cluster.xml`: For building a clustered Glassfish environment.

We know the build has been successful when we see the message `'BUILD SUCCESSFUL Total time: XX seconds'` followed by a return of the root prompt.

Note down the following port numbers which will be required later:

Admin Console	4848
HTTP Instance	8080
JMS	7676
IIOP	3700
HTTP_SSL	8181
IIOP_SSL	3820
IIOP_MUTUALAUTH	3920
JMX_ADMIN	8686

As observed, the deleted directories, `bin/`, `config/` and `domains/`, are recreated after a successful build. Verify if the build Glassfish application server supports clusters.

The reference command is:

```
# cd /appinstall/glassfish/bin
# ./asadmin
```

The build Glassfish application server supports clusters if the default domain 'domain1' starts successfully with the following message:

```
asadmin> start-domain
```

Domain supports application server clusters and other standalone instances.
asadmin>



Note: The default credentials for 'Glassfish Administrative Console' are:

Username	admin
Password	adminadmin

Cluster configuration on GF-NODE 01

Before proceeding, just ensure that the domain is running.

There are three major components in creating a cluster:

- Creating a 'node agent' for both nodes.
 - Creating a 'cluster' that will be common for both nodes - 'GF-Node01' and 'GF-Node02'.
 - Creating 'instances' for both nodes.
- Create the node agent 'GF-Agent01' for the first node 'GF-Node01' as follows:

```
asadmin> create-node-agent --host GF-Node01 --port 4848 GF-
Agent01
Please enter the admin user name=admin
Please enter the admin password=adminadmin
Command create-node-agent executed successfully.
asadmin>
```

Create the cluster 'GF-Cluster' that's common for both nodes 'GF-Node01' and 'GF-Node02' as follows:

```
asadmin> create-cluster --host GF-Node01 --port 4848 GF-Cluster
Please enter the admin user name=admin
Please enter the admin password=adminadmin
Command create-cluster executed successfully.
asadmin>
```



Figure 1: JDBC/TimerPool Error

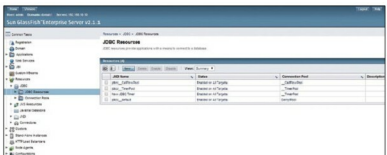


Figure 2: JDBC resources